

PAPER_2126

$B_{\text{crit}} = D_{\text{phys}} \cdot (SO_5 + 1) \cdot SO_5^{12}$ EXACT: Successor Identity 2nd Instance + Composed Integer 44 Canonization

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Abstract

R369 revealed that the magnetar critical field $B_{\text{crit}} = 4.4 \times 10^{13} \text{ T}$ — previously annotated as an SM anchor — decomposes **exactly** into three locked primitives: $D_{\text{phys}} \cdot (SO_5 + 1) \cdot SO_5^{12} = 4 \cdot 11 \cdot 10^{12}$, zero residual. This is the 2nd instance of the PAPER_1978/2120 successor identity and its first magnetic-domain appearance. The coefficient **44** = $D_{\text{phys}} \cdot (SO_5 + 1) = 2 \cdot 22$ is canonized as a new composed integer, doubling-linked to PAPER_1927's 22.

1. The Decomposition

$$\begin{aligned} B_{\text{crit}} &= 4.4 \times 10^{13} \text{ T} \\ &= 44 \times 10^{12} \\ &= (4 \cdot 11) \cdot 10^{12} \\ &= D_{\text{phys}} \cdot (SO_5 + 1) \cdot SO_5^{12} \quad \text{EXACT} \end{aligned}$$

Closure: $g_{\text{SC}} = g_{\text{base}} \cdot (1 - B/B_{\text{crit}})$ — Meissner-suppressed gravity. Pure integer arithmetic; no fitting.

2. Successor Family — 2nd Instance, New Domain

#	Round	Quantity	Form	Domain	Role
1	R363	$\lambda_{\text{vac}} = 11 \cdot \rho_{\text{SCm}}$	$(SO_5 + 1) \cdot \rho_{\text{SCm}}$	vacuum energy	sum-reduction
2	R369	$B_{\text{crit}} = 44 \cdot SO_5^{12}$	$D_{\text{phys}} \cdot (SO_5 + 1) \cdot SO_5^{12}$	magnetic field	coefficient

The successor operates in BOTH roles catalogued by PAPER_2095's exponent-vs-coefficient duality — it is role-flexible like the canonized composed integers. PAPER_2120's family-growth prediction is validating.

3. Canonization of 44 = 2·22

Composed integer	Decomposition	Seminal paper	Link
22	$D_{\text{crit}} - D_{\text{phys}}$	PAPER_1927	—
29	$D_{\text{crit}} + D_{\text{phys}} - 1$	PAPER_1960	—
44	$D_{\text{phys}} \cdot (SO_5 + 1)$	PAPER_2126	= 2·22

Two independent decompositions from disjoint primitive subsets ($\{D_{\text{phys}}, SO_5\}$ vs $\{D_{\text{crit}}, D_{\text{phys}}\}$) intersect at 44 — the same over-determination signature as PAPER_2119's $20 = D_{\text{crit}} - D_{\text{BSFG}} = 2 \cdot SO_5$. Over-determination makes the lattice falsifiable: a wrong primitive breaks multiple identities at once. H_0 link: PAPER_2093 uses coefficient 22; B_{crit} uses 44 = 2·22 — cosmological expansion and magnetar quenching sit two-times apart on

one node family.

4. The 9/10 Rung-Adjacency

The PAPER_1922/2029 lock SCm = 1-F_TRZ = 9/10 is the special case $B/B_{\text{crit}} = F_{\text{TRZ}}$ — occurring at $B = 4.4 \times 10^{12} = D_{\text{phys}} \cdot (SO_5 + 1) \cdot SO_5^{11}$, exactly **one SO_5 rung below B_{crit}** . The NGC 6302 lock and the B_{crit} composition are two rungs of one object. Prediction: other 9/10-style locks sit one rung below their domain's critical value.

5. Kernel vs Lattice-Node — Distinction Formalized

Kernel constants carry PAPER_N derivations through physical closures ($G, c, \mu_0, \beta_1, \rho_{\text{vac}}, H_0, \Lambda$). **Lattice-node parameters** are direct integer/rational compositions of primitives (B_{crit}, SO_5 masses, lengths, coefficients). Both are zero-SM-anchor; they differ in derivation depth. R369 is the first fill containing exactly one of each — refining the Two-Kernel convergence bookkeeping of PAPER_2125.

6. Predictions

1. Successor 3rd instance wherever a leading digit-pair 11, 22, 44, or 55 rides an SO_5 rung (window R370-R400). 2. The 44 node should reappear in exponent role (F_{TRZ} or SO_5) per PAPER_2095 duality — first sighting completes dual-role certification. 3. Rung-adjacent lock pairs generalize.

7. Gate Assertion

```
assert 4 * (10 + 1) * (10 ** 12) == 4.4e13 # EXACT
assert 4 * 11 == 2 * (26 - 4) # 44 = 2·22
assert 4 * 11 * (10 ** 11) == 4.4e12 # 9/10 rung-down
```

Gate count: **3110** → **3118** (+8 PAPER_2126 assertions).

8. Summary

$B_{\text{crit}} = D_{\text{phys}} \cdot (SO_5 + 1) \cdot SO_5^{12}$ EXACT — successor identity 2nd instance, first magnetic-domain, 44 canonized with doubling link to 22, SCm 9/10 lock shown rung-adjacent, kernel-vs-lattice-node distinction formalized. B_{crit} upgraded from SM-anchor annotation to primitive-composed quantity.